A Novel Federal Learning-based Framework Defending Server-Level Privacy Leakage

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Abstract—The development of Internet of Things and the prosperity of 5G communication technologies generate enormous data that promotes the rapid progress of artificial intelligence applications. However, with the progress of social civilization and the increment of the well-defined laws, the emphasis on the ownership and security of data as well as the user privacy have become a worldwide major issue that challenges the centralized computing. Although existing federated learning-based studies have been fruitful, they still face the challenges from inference attacks and reconstruction attacks. Therefore, we propose a novel security aggregation mechanism of confusion gradients (namely SAM) to withstand the inference and reconstruction attacks that against federated learning-based systems, and build a novel framework (namely D-FLSPL) to perform SAM for server-level privacy leakage defense. Our experimental results on three real-world datasets show that D-FLSPL performs well, which demonstrates the correctness and the security of our proposed SAM.

Index Terms—Confusion Gradient, Gradient Leakage, Privacy-Preserving, Federal Learning, Internet of Things.

1 Introduction

The development of Internet of Things (IoTs) and the prosperity of 5G communication technologies, not only bring revolutionary changes to people's daily lives, but also generate enormous data that promotes the rapid progress of artificial intelligence (AI) application models, such as intelligent transportation systems [17, 27], and recommendation systems [8, 26, 29], etc. For the multi-source data with multi-topic, multi-terminal, multi-industry and multi-platform, the traditional way of training and constructing an AI model is the centralized computing, where all data generated by clients are uploaded to a central point that owns high-performance computing cluster capable of training and running AI models. In the past decade, many methods have been proposed to achieve the performance improvement of the centralized computing [12]. However, with the progress of social civilization and the increment of the well-defined laws and regulations, two disadvantages limit the development of centralized computing studies. On the one hand, uploading the data to a central point means that data owners lose their ownerships of the data. Most of data owners heavily resist uploading their data because of industry competition. On the other hand, users are becoming more concerned about whether their privacy information is being used for commercial purposes without their consents. That is, the emphasis on the ownership of data as well as the user privacy have become a worldwide major issue that challenges the centralized computing-based methods.

Fortunately, a new approach, which is known as Federated Learning (FL), becomes an effective way dealing with the above challenge because of its characters of data controllability, training cooperativity and virtual openness. The data controllability mainly represents that the data generated by each client will not leave the current local area and meanwhile the ownership of the data will not be transferred and shared. The training cooperativity is mainly manifested in that for each client, there is a local AI model trained according to the local data, and then a global model is constructed by aggregating the local AI models. During the process of constructing the global model, the data generated by all clients seems to be integrated together, which is called virtual openness. In addition, in order to further guarantee the security and privacy of the data, FL-based studies utilize various privacy techniques, such as secure multiparty computation (SMC) [22], differential privacy [20], etc. However, a FL system is vulnerable to inference attacks and reconstruction attacks if a curious client that can work with the curious central server appears, thus resulting in data leakage [2].

In a FL system, the global model is essentially learned according to the parameters uploaded by clients. The parameters contain the properties of local data so that the local data can be recovered if a corresponding decoder could be constructed in reverse [24, 30]. Although various privacy techniques are utilized to guarantee the security of the transferring parameters, the privacy information of transferring parameters could be still exposed to the curious central server if a curious client that can be cooperated with appears. Therefore, we propose a novel security aggregation mechanism of confusion gradients (SAM in short) to withstand the inference attacks and reconstruction attacks that against FL systems, by confusing the transferring parameters of clients with each other. In addition, we build a novel framework, namely D-FLSPL, to perform SAM for server-level privacy leakage defense.

The main contributions are listed as follows.

- We propose a novel federated learning-based framework defending server-level privacy leakage, namely D-FLSPL. By utilizing the characters of FL, such as data controllability, training cooperativity and virtual openness, D-FLSPL can not only guarantee the data ownerships of data owners and the secure of user privacy, but also effectively withstand the inference attacks and reconstruction attacks that against FL systems.
- We propose a novel security aggregation mechanism of confusion gradients, namely SAM. SAM first splits the full gradient of each client into two incomplete ones with different sizes, and then randomly send an incomplete gradient to a randomly-selected client to perform confusion operations. After that, all gradients that clients send to the central server are different from each other in terms of sizes and types. Note that SAM runs in each round of iteration during the process of learning a global model.
- We conduct multiple groups of experiments on three real-world datasets (i.e., $MNIST$, $CIFAR$-10 and $CIFAR$-100) to evaluate our D-FLSPL and SAM. Our experimental results show that D-FLSPL and SAM perform well in terms of correctness and the security. This also demonstrates the effectiveness and the reasonableness of confusing gradients.

The remainder of the paper is organized as follows. Section 2 introduces our motivations. Section 3 presents our proposed framework D-FLSPL. Section 4 reports experimental results and corresponding analyses. Section 5 summarizes related works. And finally, we conclude the paper in Section 6.

2 Motivations

In this section, we will particularly explain our motivations: i) why is federated learning introduced? And ii) why is the security aggregation mechanism of confusion gradients introduced?

2.1 Why Introducing Federated Learning

Generally speaking, training an AI model requires very large volume of data. Intuitively, an AI model could be run locally if enormous and adequate data is available in the local area. But only little data can be locally obtained in many industries in reality. That is, the obtained data is either small in terms of volume or missing important labels and characteristic values [25]. Therefore, moving local data into a cloud center is an effective way to increase the amount of the adequate data, so as to improve the quality of training AI models by centralized computing [13]. However, in the era of big data, for the explosive increment data with multi-source (such as multi-topic, multi-terminal, multi-industry and multi-platform), the increased data traffic, the well-defined laws, the awareness of preventing privacies and the ownership of data challenge the centralized computing, as shown in Fig. 1.

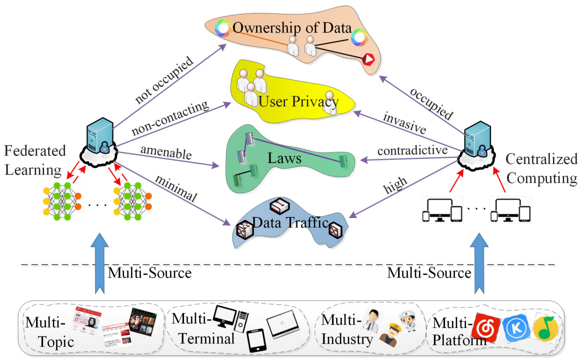


Fig. 1. The comparisons between centralized computing and federated learning.

Federated learning can effectively deal with these issues that the centralized computing faces, also as shown in Fig. 1. In federated learning systems, although there are interactions between the central server and the clients, the data transferred between them only contains the gradients (parameters) that the AI models require. That is, compared with the centralized computing, federated learning only requires minimal data traffic since the local data do not have to be moved into the central server. Also because of such, the central server in federated learning do not contact or take the ownership of local data, thus protecting the security and the ownership of local data as well as the user privacy, and meanwhile not violating the well-defined laws. However, federated learning systems face an inevitable challenge from inference attacks and reconstruction attacks since the gradients transferred between the central server and clients are full. Therefore, this paper introduces a security aggregation mechanism of confusion gradients, i.e., Motivation Two.

2.2 Why Introducing Security Aggregation Mechanism of Confusion Gradients

Because the gradients transferred between the central server and each client are full, by cooperating with a curious client, the curious central server could retrieve the parameters of clients using inference attacks and reconstruction attacks, thus recovering their local data. The security aggregation mechanism of confusion gradients aims to prevent inference attacks and reconstruction attacks, by changing the independent full gradients into confusion ones. Fig. 2 shows a scenario of running the security aggregation mechanism of confusion gradients. The global model is deployed in the central server, and LM_1, LM_2 and LM_3 denote three independent local models respectively deployed in the three individual clients.

In the t-th update operation, first, the global model respectively send LM_1, LM_2 and LM_3 an initial gradient $\triangledown \boldsymbol{W_t}$, and LM_1, LM_2 and LM_3 respectively obtain their local initial gradients

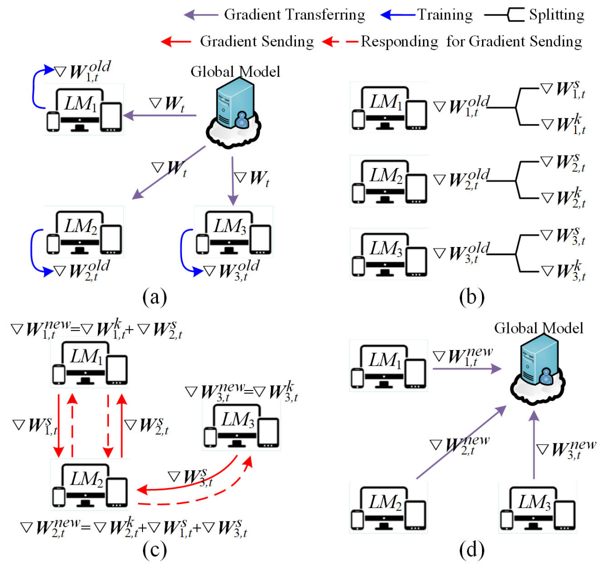


Fig. 2. An example of explaining the security aggregation mechanism of confusion gradients.

$\triangledown \boldsymbol{W_{1,t}^{old}}$, $\triangledown \boldsymbol{W_{2,t}^{old}}$ and $\triangledown \boldsymbol{W_{3,t}^{old}}$ by training $\triangledown \boldsymbol{W_t}$, as shown in Fig. 2(a). Note that these three local initial gradients are full. Second, LM_1 randomly splits $\triangledown \boldsymbol{W_{1,t}^{old}}$ into two incomplete parts with different sizes, i.e., a sending part $\triangledown \boldsymbol{W_{1,t}^{s}}$ and a keeping one $\triangledown \boldsymbol{W_{1,t}^{k}}$. Similarly, $\boldsymbol{W_{2,t}^{old}}$ is split into $\triangledown \boldsymbol{W_{2,t}^{s}}$ and $\triangledown \boldsymbol{W_{2,t}^{k}}$, and $\triangledown \boldsymbol{W_{3,t}^{old}}$ is split into $\triangledown \boldsymbol{W_{3,t}^{s}}$ and $\triangledown \boldsymbol{W_{3,t}^{k}}$, as shown in Fig. 2(b). Third, LM_1 sends $\triangledown \boldsymbol{W_{1,t}^{s}}$ to a randomly-selected client (i.e., LM_2); LM_2 sends $\triangledown \boldsymbol{W_{2,t}^{s}}$ to a randomly-selected client LM_1; and LM_3 sends $\triangledown \boldsymbol{W_{3,t}^{s}}$ to a randomly-selected client LM_2; see the red arrows with solid lines in Fig. 2(c). During this step, a client may be selected by N clients (except itself), where $N \in$ {0, 1, 2, 3, . . . }. After that, LM_1 obtains a confusion gradient $\triangledown \boldsymbol{W_{1,t}^{new}}$ according to incomplete gradients $\triangledown \boldsymbol{W_{1,t}^{k}}$ and $\triangledown \boldsymbol{W_{2,t}^{s}}$; LM_2 obtains a confusion gradient $\triangledown \boldsymbol{W_{2,t}^{new}}$ according to incomplete gradients $\triangledown \boldsymbol{W_{2,t}^{k}}$, $\triangledown \boldsymbol{W_{1,t}^{s}}$ and $\triangledown \boldsymbol{W_{3,t}^{s}}$; and LM_3 obtains a confusion gradient $\triangledown \boldsymbol{W_{3,t}^{new}}$ according to incomplete gradient $\triangledown \boldsymbol{W_{3,t}^{k}}$, as shown in Fig. 2(c). At last, LM_1, LM_2 and LM_3 respectively send their confusion gradients $\triangledown \boldsymbol{W_{1,t}^{new}}$, $\triangledown \boldsymbol{W_{2,t}^{new}}$ and $\triangledown \boldsymbol{W_{3,t}^{new}}$ to the global model, as shown in Fig. 2(d). In the next section, our D-FLSPL framework proposed according to the two motivations will be introduced in detail.

3 D-FLSPL Framework

In this section, we will introduce the proposed D-FLSPL framework. First, some relevant definitions of federated learning [25], inference attacks [23] and reconstruction attacks [1] that help construct our D-FLSPL will be reviewed. Then three new definitions that support our D-FLSPL will be reported. And finally, the proposed D-FLSPL will be described in detail.

3.1 Preliminaries

Definition 1 (Federated Learning, FL). *A federated learning system is a learning process where the clients collaboratively train a global model, in which process any data owner C_i does not expose its data D_i to others. Note that $C_i \in \{C_1, C_2, C_3, \ldots, C_N\}$, $D_i \in \{D_1, D_2, D_3, \ldots, D_N\}$, and N denotes a natural number.*

Definition 2 (Membership-Inference Attacks)**.** *In membership-inference attacks (inference attack), the attacker has black-box access to a model, as well as a certain sample, as its knowledge. The attacker's target is to construct an attack model that can learn if the sample is inside the training set based on the model output. The attacker conducts such attacks by finding and leveraging the differences in the model predictions on the samples belonging to the training set vs. other samples.*

Definition 3 (Reconstruction Attack)**.** *Reconstruction attacks require white-box access to AI models. That is, the feature vectors in an AI model must be known. The adversary's goal is reconstructing the raw private data by using its knowledge of the feature vectors. Such attacks could be possible when the feature vectors used for the AI model training phase were not deleted after building the desired AI model.*

In order to understand the proposed security aggregation mechanism of confusion gradients clearly and easily, here we first report three new definitions, i.e., $Full\ Gradient$, $Incomplete\ Gradient$ and $Confusion\ Gradient$, that support constructing our D-FLSPL, based on existing definitions of FL.

Definition 4 (Full Gradient)**.** *A full gradient of a local model in an update operation, which is defined as a local initial gradient of the local model in the current update operation, is obtained by training the global initial gradient coming from the central server. Note that for each local model in each update operation, there is a corresponding full gradient generated.*

For example, as shown in Fig. 1(a), in the t-th update operation, the local model LM_1 obtains a local initial gradient $\bigtriangledown \boldsymbol{W_{1,t}^{old}}$ by training the global initial gradient $\bigtriangledown \boldsymbol{W_t}$ coming from the central server. The $\bigtriangledown \boldsymbol{W_{1,t}^{old}}$ is the full gradient of LM_1 in the t-th update operation.

Definition 5 (Incomplete Gradient)**.** *An incomplete gradient of a local model in an update operation is defined as a part of the full gradient of the local model in the current update operation.*

For example, as shown in Fig. 1(b), in the t-th update operation, the full gradient of LM_1 (i.e., $\bigtriangledown \boldsymbol{W_{1,t}^{old}}$) is split into a sending part $\bigtriangledown \boldsymbol{W_{1,t}^{s}}$ and a keeping part $\bigtriangledown \boldsymbol{W_{1,t}^{k}}$. The sending part $\bigtriangledown \boldsymbol{W_{1,t}^{s}}$ and the keeping part $\bigtriangledown \boldsymbol{W_{1,t}^{k}}$ are two incomplete gradients of LM_1.

Definition 6 (Confusion Gradient)**.** *A confusion gradient of a local model in an update operation is a mixture that contains an incomplete gradient of the local model and i incomplete gradients from the other local models, where $i \in \{0, 1, 2, 3, \dots, n\}$, n denotes the number of local models in a FL system. Note that the confusion gradient of a local model is treated as the final gradient that will be sent to the global model.*

Here, we still utilize the simple example in Fig .1 to further explain what a confusion gradient is. As shown in Fig .1(c), LM_1 obtains a confusion gradient $\bigtriangledown \boldsymbol{W_{1,t}^{new}}$ containing one of its own incomplete gradients $\bigtriangledown \boldsymbol{W_{1,t}^{k}}$ and an incomplete gradient $\bigtriangledown \boldsymbol{W_{2,t}^{s}}$ from LM_2; the confusion gradient $\bigtriangledown \boldsymbol{W_{2,t}^{new}}$ of LM_2 contains one of its own incomplete gradients $\bigtriangledown \boldsymbol{W_{2,t}^{k}}$, an incomplete gradient $\bigtriangledown \boldsymbol{W_{1,t}^{s}}$ from LM_1 and an incomplete gradient $\bigtriangledown \boldsymbol{W_{3,t}^{s}}$ from LM_3; and the confusion gradient $\bigtriangledown \boldsymbol{W_{3,t}^{new}}$ of LM_3 only contains one of its own incomplete gradients $\bigtriangledown \boldsymbol{W_{3,t}^{k}}$.

After reviewing the relevant definitions that support constructing the proposed D-FLSPL, in the following paragraphs, we will first describe the overview of our D-FLSPL, and then report the security and correctness verifications of the secure aggregation mechanism of confusion gradients (SAM in short).

Algorithm 1 The main pseudo-code of SAM of the i-th local model in the t-th update operation.

Input: A global initial gradient $\bigtriangledown \boldsymbol{W_t}$; an communication address list L^{CA}.
Output: The confusion gradient $\bigtriangledown \boldsymbol{W_{i,t}^{new}}$.

$LM_i.init(\bigtriangledown \boldsymbol{W_t}$, $L^{CA})$;
if $sizeof(L^{CA}) \leq 1 \parallel empty(\bigtriangledown \boldsymbol{W_t})$==$\boldsymbol{true}$ **then**
  return ;
end if
CA_{GM}=L^{CA}.get(0);
L^{CA}.remove(0);
$LM_i.receiveMonitoring()$;
$\bigtriangledown \boldsymbol{W_{i,t}^{old}}$ = $LM_i.localTraining(\bigtriangledown \boldsymbol{W_t})$;
$Array_{i,t}$=$LM_i.splitTwoParts(\bigtriangledown \boldsymbol{W_{i,t}^{old}})$;
$\bigtriangledown \boldsymbol{W_{i,t}^{s}}$=$Array_{i,t}$.get(0);
$\bigtriangledown \boldsymbol{W_{i,t}^{k}}$=$Array_{i,t}$.get(1);
while LM_i.responded==$\boldsymbol{false}$ **do**
  $LM_i.send$(0, L^{CA}, $\bigtriangledown \boldsymbol{W_{i,t}^{s}}$);
  $sleep$(10000);
end while
$sleep(LM_i.ScheduleTime)$;
$LM_i.endMonitoring()$;
$\bigtriangledown \boldsymbol{W_{i,t}^{new}}$=$\bigtriangledown \boldsymbol{W_{i,t}^{k}}$;
for each $x \in \{$0, 1, 2, 3, ..., L^{W}.length-1$\}$ **do**
  $\bigtriangledown \boldsymbol{W_{i,t}^{new}}$+=$L^{CA}$.get($x$);
end for
$LM_i.Transfer(CA_{GM}$, $\bigtriangledown \boldsymbol{W_{i,t}^{s}})$;

Algorithm 2 The function receiveMonitoring.

while LM_i.receiving **do**
  if LM_i.receiveType==0 **then**
    L^{W}.append(LM_i.getIncompeteGradient);
    $RespondAddress$=LM_i.receiveAddress;
    $ResondValue$=$\boldsymbol{true}$;
    $LM_i.send$(1, $RespondAddress$, $ResondValue$);
  else if LM_i.receiveType==1 **then**
    LM_i.responded=$ResondValue$;
  end if
end while

3.2 The Overview of D-FLSPL Framework

Specifically, D-FLSPL consists of a central server that deploys a global model and N clients, where each individual client deploys an independent local model. The architecture of D-FLSPL framework is shown in Fig. 3. Particularly, D-FLSPL takes nine steps to perform an update operation, in which Steps 2 to 7 are the key content of the proposed SAM. Here we only take the i-th local model (denoted as LM_i) that runs in the t-th update operation as an example to explain how the proposed D-FLSPL and SAM work, since the processes of local models are the same. Note that the lines without algorithm name appearing in the steps are shown in *Algorithm 1*.

- Step 1. The global model (denoted as M_{FED}) first confirms that all clients are online, and then send all clients the

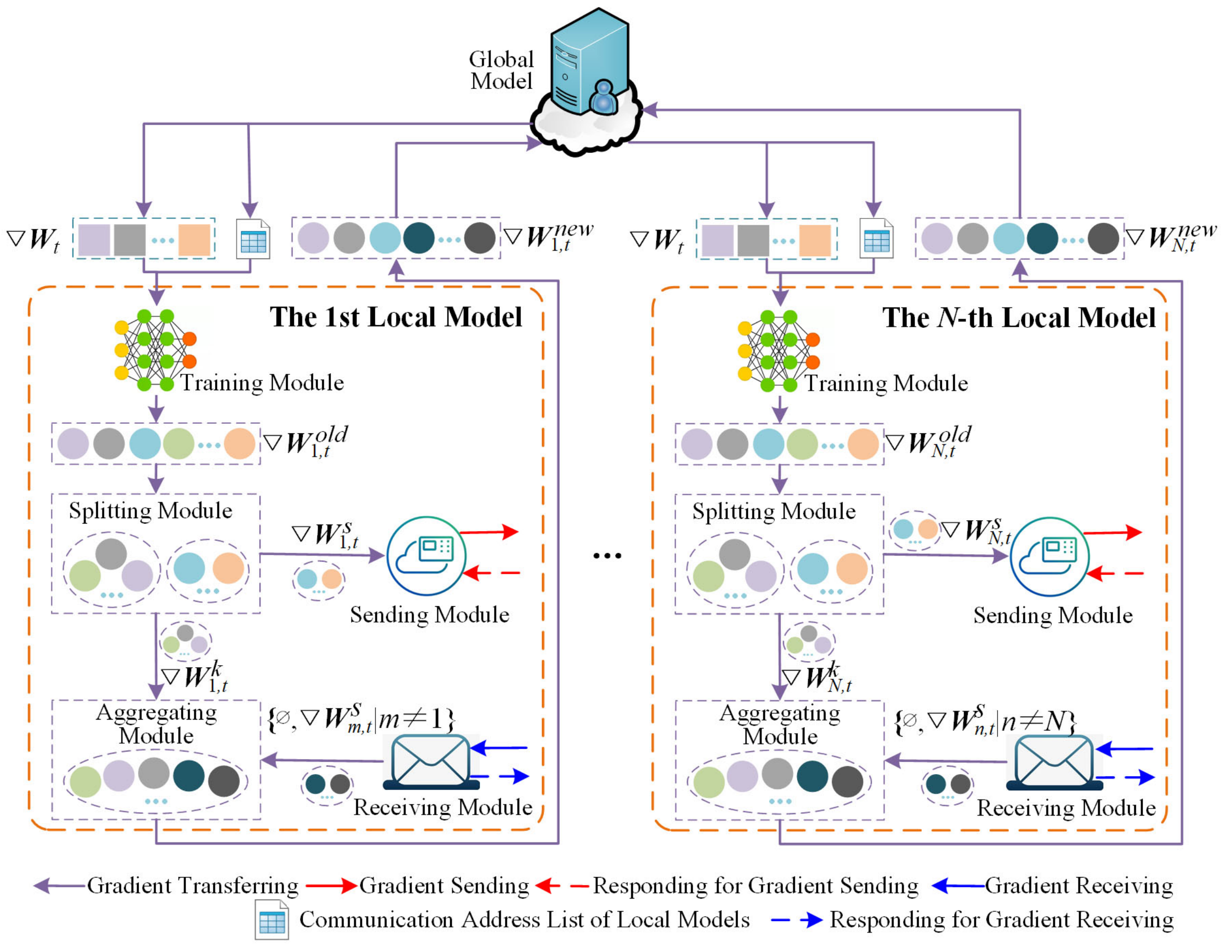


Fig. 3. The architecture of D-FLSPL framework.

global initial gradient $\bigtriangledown \boldsymbol{W_t}$ and a communication address list L^{CA}.

- Step 2. If $\bigtriangledown \boldsymbol{W_t}$ is empty or the size of L^{CA} is lower than one, LM_i stops running, as shown in Lines 1 to 4. Otherwise, LM_i extracts the communication address of the global model from L^{CA}, denoted as CA_{GM} (see Line 5), and removes it from L^{CA} (see Line 6). Note that L^{CA} does not contain the communication address of the current local model. That is, L^{CA} does not contain the communication address of LM_i.
- Step 3. LM_i activates a monitor module (see Line 7). The monitor module runs until the monitor ending module is called (see Line 17). The monitor module performs two functionalities, where i) it receives incomplete gradients sent by the other local models and sends feedbacks to the senders (see Lines 2 to 6 in *Algorithm 2*); and ii) it receives the feedback from the local model receiving the incomplete gradient that LM_i sent (see Lines 7 to 8 in *Algorithm 2*).
- Step 4. LM_i trains the local model according to local data and $\bigtriangledown \boldsymbol{W_t}$ to obtain a local initial gradient $\bigtriangledown \boldsymbol{W_{i,t}^{old}}$, and then randomly splits $\bigtriangledown \boldsymbol{W_{i,t}^{old}}$ into two parts with different sizes, i.e., a sending part $\bigtriangledown \boldsymbol{W_{i,t}^{s}}$ and a keeping one $\bigtriangledown \boldsymbol{W_{i,t}^{k}}$, see Lines 8 to 11. Note that $\bigtriangledown \boldsymbol{W_{i,t}^{old}}$ is an independent full gradient of LM_i.
- Step 5. LM_i sends the sending part $\bigtriangledown \boldsymbol{W_{i,t}^{s}}$ to a randomly-selected local model LM_n. The communication address of LM_n can be find from L^{CA}. Note that LM_i periodically sends $\bigtriangledown \boldsymbol{W_{i,t}^{s}}$ until a feedback from LM_n is received. In this paper, the periodicity is assigned 10 seconds (see Lines 12 to 15).
- Step 6. LM_i waits for a period of time (denoted as $ScheduleTime$) until all local models finished sending their sending parts, and then ends the monitor module (see Line 16 to 17), since i) local models require different processing time because of different hardware and software environments; and ii) a local model may be selected by N local models, where $N \in \{0, 1, 2, 3, \dots\}$.
- Step 7. LM_i integrates its keeping gradient $\bigtriangledown \boldsymbol{W_{i,t}^{k}}$ and incomplete gradients from the other local models into a confusion gradient $\bigtriangledown \boldsymbol{W_{1,t}^{new}}$, and then sends $\bigtriangledown \boldsymbol{W_{1,t}^{new}}$ to M_{FED}, see Lines 18 to 22.
- Step 8. A new global initial gradient $\bigtriangledown \boldsymbol{W_{t+1}}$ is obtained by aggregating the confusion gradients sent by local models, and then is utilized to retrain M_{FED}.
- Step 9. If i) M_{FED} converges; ii) the maximum number of iterations is reached; or iii) the maximum training duration is reached, D-FLSPL stops running. Otherwise, go to Step 1.

After finishing the nine steps, the central server obtains an updated global model M_{FED}. Finally, all clients re-download M_{FED} to redeploy.

3.3 Security and Correctness Verifications of SAM

The SAM is the key content of our proposed D-FLSPL. In this subsection, we will demonstrate the effectiveness and the reasonableness of the proposed SAM respectively in terms of the security verification and correctness verification.

Correctness Verification. For the t-th update operation in a FL system, the central server aggregates gradients sent by local models according to:

$$\triangledown \boldsymbol{W_{t+1}} = \frac{1}{N}\sum_{i=1}^{N}\triangledown \boldsymbol{W_{i,t}} \tag{1}$$

where $\triangledown \boldsymbol{W_{t+1}}$ denotes the global initial gradient of the (t+1)-th update operation, N denotes the number of clients in the FL system, and the $\triangledown \boldsymbol{W_{i,t}}$ denotes the local gradient sent by the i local model in the t-th update operation. Thus, in this paper, the global initial gradient of the (t+1)-th update operation is calculated as follows, without employing SAM:

$$\triangledown \boldsymbol{W_{t+1}} = \frac{1}{N}\sum_{i=1}^{N}\triangledown \boldsymbol{W_{i,t}^{old}} \tag{2}$$

According to definitions listed in Subsection 3.1 and statements in Subsection 3.2, we know that SAM does not append or discard any incomplete gradients (i.e., the sending one $\triangledown \boldsymbol{W_{i,t}^{s}}$ and the keeping one $\triangledown \boldsymbol{W_{i,t}^{k}}$) that derived from the local initial gradients. That is, SAM only changes the senders of the sending ones. Thus, the global initial gradient of the (t+1)-th update operation is calculated as follows while employing SAM:

$$\begin{aligned}\triangledown \boldsymbol{W_{t+1}} &= \frac{1}{N}\sum_{i=1}^{N}\triangledown \boldsymbol{W_{i,t}^{new}}\\ &= \frac{1}{N}(\sum_{i=1}^{N}\triangledown \boldsymbol{W_{i,t}^{s}} + \sum_{i=1}^{N}\triangledown \boldsymbol{W_{i,t}^{k}})\\ &= \frac{1}{N}\sum_{i=1}^{N}(\triangledown \boldsymbol{W_{i,t}^{s}} + \triangledown \boldsymbol{W_{i,t}^{k}})\\ &= \frac{1}{N}\sum_{i=1}^{N}\triangledown \boldsymbol{W_{i,t}^{old}}\end{aligned} \tag{3}$$

where $\triangledown \boldsymbol{W_{i,t}^{new}}$ denotes the confusion gradient of the i local model in the t-th update operation. The aggregation result that Eq. (3) figures out is the same as that calculated by Eq. (2), which demonstrates that the reasonableness of the proposed SAM in terms of correctness verification.

Here, we utilize the example in Fig. 2 to explain the correctness verification of our SAM. The initial gradients (i.e., $\triangledown \boldsymbol{W_{1,t}^{old}}$, $\triangledown \boldsymbol{W_{2,t}^{old}}$ and $\triangledown \boldsymbol{W_{3,t}^{old}}$) and the confusion gradients (i.e., $\triangledown \boldsymbol{W_{1,t}^{new}}$, $\triangledown \boldsymbol{W_{2,t}^{new}}$ and $\triangledown \boldsymbol{W_{3,t}^{new}}$) of three local models in the t-th update operation are listed as follows:

LM_1:
$\triangledown \boldsymbol{W_{1,t}^{old}}$=$\triangledown \boldsymbol{W_{1,t}^{s}}$+$\triangledown \boldsymbol{W_{1,t}^{k}}$; $\triangledown \boldsymbol{W_{1,t}^{new}}$=$\triangledown \boldsymbol{W_{1,t}^{k}}$+$\triangledown \boldsymbol{W_{2,t}^{s}}$
LM_2:
$\triangledown \boldsymbol{W_{2,t}^{old}}$=$\triangledown \boldsymbol{W_{2,t}^{s}}$+$\triangledown \boldsymbol{W_{2,t}^{k}}$; $\triangledown \boldsymbol{W_{2,t}^{new}}$=$\triangledown \boldsymbol{W_{2,t}^{k}}$+$\triangledown \boldsymbol{W_{1,t}^{s}}$+$\triangledown \boldsymbol{W_{3,t}^{s}}$
LM_3:
$\triangledown \boldsymbol{W_{3,t}^{old}}$=$\triangledown \boldsymbol{W_{3,t}^{s}}$+$\triangledown \boldsymbol{W_{3,t}^{k}}$; $\triangledown \boldsymbol{W_{3,t}^{new}}$=$\triangledown \boldsymbol{W_{3,t}^{k}}$

The global initial gradient of the (t+1)-th update operation is calculated as follows:

$$\begin{aligned}\triangledown \boldsymbol{W_{t+1}} &= \frac{1}{N}(\triangledown \boldsymbol{W_{1,t}^{new}} + \triangledown \boldsymbol{W_{2,t}^{new}} + \triangledown \boldsymbol{W_{3,t}^{new}})\\ &= \frac{1}{N}[(\triangledown \boldsymbol{W_{1,t}^{k}} + \triangledown \boldsymbol{W_{2,t}^{s}}) + (\triangledown \boldsymbol{W_{2,t}^{k}}\\ &\quad + \triangledown \boldsymbol{W_{1,t}^{s}} + \triangledown \boldsymbol{W_{3,t}^{s}}) + (\triangledown \boldsymbol{W_{3,t}^{k}})]\\ &= \frac{1}{N}[(\triangledown \boldsymbol{W_{1,t}^{s}} + \triangledown \boldsymbol{W_{1,t}^{k}})\\ &\quad + (\triangledown \boldsymbol{W_{2,t}^{s}} + \triangledown \boldsymbol{W_{2,t}^{k}})\\ &\quad + (\triangledown \boldsymbol{W_{3,t}^{s}} + \triangledown \boldsymbol{W_{3,t}^{k}})]\\ &= \frac{1}{N}(\triangledown \boldsymbol{W_{1,t}^{old}} + \triangledown \boldsymbol{W_{2,t}^{old}} + \triangledown \boldsymbol{W_{3,t}^{old}})\end{aligned} \tag{4}$$

From Eq. (4) we can find that the aggregation result computed by the confusion gradients of local models is the same as that calculated according to the initial gradients of local models. This demonstrates the reasonableness of SAM.

Security Verification. In a FL system with closed-cycle, inference attacks and reconstruction attacks are mainly launched by the curious central server. By cooperating with a curious client, the curious central server could retrieve the parameters that clients utilize, thus recovering the local data of clients, since the gradients transferred between the curious central server and each client hold the parameter completeness. SAM aims to break the parameter completeness.

According to the statements in Subsection 3.2, we know that SAM works in each round of update operation, and after breaking the parameter completeness in an update operation, the confusion gradient that a local model LM_n sends to the global model may i) only contain the keeping gradient of LM_n, since LM_n is not selected by any other local models; ii) contain the keeping gradient of LM_n and one or more sending gradients coming from other local models, since LM_n is selected by one or more other local models. That is, i) in an update operation, the confusion gradient that a local model sends to the global model is different from the other local models in terms of the size and the type of parameters; and ii) in each update operation, the parameters that the confusion gradient of a local model contains may be different. Therefore, no matter which local model is a curious client that can be cooperated with, the curious central server cannot retrieve the parameters of each local model using inference attacks or recover the local data of clients utilizing reconstruction attacks.

For example, as shown in Fig. 2(a), the local initial gradients $\triangledown \boldsymbol{W_{1,t}^{old}}$, $\triangledown \boldsymbol{W_{2,t}^{old}}$ and $\triangledown \boldsymbol{W_{3,t}^{old}}$ that LM_1, LM_2 and LM_3 obtain by training the global initial gradient $\triangledown \boldsymbol{W_t}$ hold the completeness of parameters. Hence, if LM_1 is a curious client, the curious central server could retrieve the parameters of LM_2 and LM_3 to obtain their local data, by cooperating with LM_1. However, the confusion gradients $\triangledown \boldsymbol{W_{1,t}^{new}}$, $\triangledown \boldsymbol{W_{2,t}^{new}}$ and $\triangledown \boldsymbol{W_{3,t}^{new}}$ obtained by SAM break the parameter completeness, so that $\triangledown \boldsymbol{W_{1,t}^{new}}$, $\triangledown \boldsymbol{W_{2,t}^{new}}$ and $\triangledown \boldsymbol{W_{3,t}^{new}}$ are different from each other in terms of sizes, as shown in Fig. 2(c). Thus, the global model cannot recover the parameters and the local data of clients by cooperating with any curious client.

In addition, since the proposed framework D-FLSPL performs SAM in an update operation, any two clients require twice or more communications to achieve the confusion operations of incomplete gradients, which inevitably increases the overhead of time and bandwidth. However, with the prosperity of mobile

Internet, such as 4G and 5G networks, the extra overhead of time and bandwidth can be ignored. Besides, if a client is offline and can not be back online within a specified time in an update operation, the corresponding results of gradient aggregations in the current update operation will be discarded.

4 Experiments

In this section, we will first introduce datasets used in our experiments, and then we will report our experimental results in terms of correctness verification and security verification.

4.1 Datasets

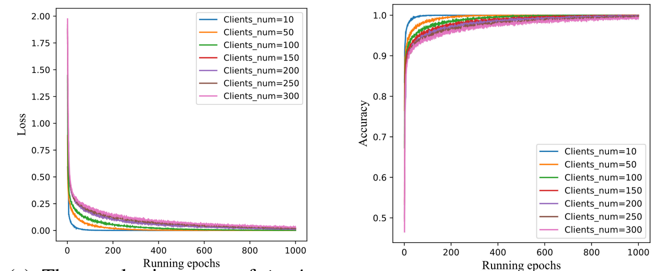
The experiments are conducted on three real-world datasets that are widely utilized in many research fields, which are listed as follows:

- $MNIST$ [LeCun et al.]. $MNIST$, which is divided into a training set and a test set, contains 70,000 samples of handwritten digits from 0 to 9, where each sample is a grayscale image of a handwritten digit with size of 28*28*1 pixel. The training set contains 60,000 samples and the test set contains 10,000 samples.
- $CIFAR$-10 [Krizhevsky et al.]. $CIFAR$-10 contains 60,000 32*32 color images (samples) in 10 classes, where 50,000 for training and 10,000 for testing. Each class contains 6,000 samples. $CIFAR$-10 is divided into five training batches and one test batch, where the test batch includes exactly 1,000 samples that are randomly selected from each class and each training batch includes the remaining samples in random order.
- $CIFAR$-100 [Krizhevsky et al.]. $CIFAR$-100 contains 60,000 32*32 color images (samples) in 100 classes, where 50,000 for training and 10,000 for testing. Each class contains 600 samples, where 500 for training and 100 for testing. The 100 classes that $CIFAR$-100 are grouped into 20 superclasses.

4.2 Experimental Results on Correctness Verification

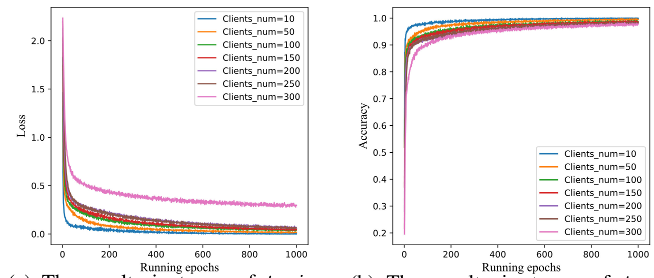
In the Subsection 3.3, the correctness verification has been discussed. Here, three groups of experiments are conducted on three datasets to verify the correctness of the proposed D-FLSPL. The three datasets are respectively $MNIST$, $CIFAR$-10 and N-IDD-$MNIST$. The N-IDD-$MNIST$ dataset is obtained based on $MNIST$ via some particular operations. In the real world, the data that local models utilize in a federated learning system are normally not independent identically distributed whereas the $MNIST$ is an independent identically distributed dataset. Therefore, we turn the $MNIST$ into a none-independent and identically distributed dataset, denoted as N-IDD-$MNIST$, by sorting the samples according to their numerical labels. For the datasets $MNIST$, N-IDD-$MNIST$ and $CIFAR$-10, all samples that they contain are split into $Clients\_num$ fragments, where $Clients\_num$ denotes the number of clients. After that, each client is assigned a fragment. The corresponding experimental results are shown in Fig. 4, Fig. 5 and Fig. 6.

Fig. 4 shows the experimental results of D-FLSPL on $MNIST$ under different numbers of clients in terms of $train\ loss$ and $test\ accuracy$. As shown in Fig. 4(a), with the increment of the FL's iterations, the value of $train\ loss$ is getting smaller. In addition, with the increment of the FL's iterations, the



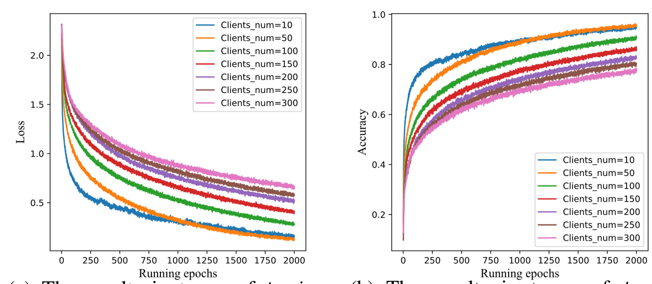
(a) The results in terms of $train\ loss$ (b) The results in terms of $test\ accuracy$

Fig. 4. The experimental results of D-FLSPL on $MNIST$ under different numbers of clients.



(a) The results in terms of $train\ loss$ (b) The results in terms of $test\ accuracy$

Fig. 5. The experimental results of D-FLSPL on N-IDD-$MNIST$ under different numbers of clients.



(a) The results in terms of $train\ loss$ (b) The results in terms of $test\ accuracy$

Fig. 6. The experimental results of D-FLSPL on $CIFAR$-10 under different numbers of clients.

value of $test\ accuracy$ is getting more accurate, as shown in Fig. 4(b). Note that the value of $train\ loss$ is infinitely close to 0 and the value of $test\ accuracy$ is infinitely close to 1 when the number of the iterations reach 1,000. This demonstrates that the model basically converges with excellent performance. However, Fig. 4 also shows that with the increment of the number of clients, the value of $train\ loss$ increases a little and the value of $test\ accuracy$ slightly decreases.

Fig. 5 shows the experimental results of D-FLSPL on N-IDD-$MNIST$ under different numbers of clients in terms of $train\ loss$ and $test\ accuracy$. The experimental results in Fig. 5 are similar to those shown in Fig. 4, except the result with $Clients\_num$=300 in terms of $train\ loss$, as shown in Fig. 5(a).

Compared with the grayscale images in $MNIST$ and N-IDD-$MNIST$, the color images that $CIFAR$-10 contains are relatively more complex and difficult to accurately identify. That is why another group of experiments is conducted on $CIFAR$-10. From Fig. 6 we can find that, with the increment of the FL's iterations, the value of $train\ loss$ is getting smaller (as shown in Fig. 6(a)), and the value of $test\ accuracy$ is getting more accurate (as shown in Fig. 6(b)). Note that all curves that denote the values of $train\ loss$ and $test\ accuracy$ under different numbers of clients flatten out when the number of the iterations reach 2,000, which demonstrate again that the model basically converges with excellent performance.

In addition, in order to further verify the correctness of the proposed D-FLSPL, an original FL system (denoted as O-FL) is selected as the baseline, and two more groups of comparison experiments are conducted on the three datasets (i.e., $MNIST$, $CIFAR$-10 and N-IDD-$MNIST$) under different numbers of clients in terms of $train\ loss$ and $test\ accuracy$. Note that the experimental environments that O-FL and D-FLSPL run as well as the corresponding experimental settings are the same. The corresponding comparisons in terms of $train\ loss$ and $test\ accuracy$ are respectively shown in Table 1 and Table 2.

From Table 1 and Table 2 we find that the differences between O-FL and D-FLSPL in terms of $train\ loss$ and $test\ accuracy$ are so small that can be neglected. These compared results demonstrate that the proposed D-FLSPL performs well. To sum up, we can conclude that the proposed D-FLSPL passes the correctness verification since it performs excellent not only on the independent identically distributed datasets (i.e., $MNIST$ and $CIFAR$-10), but also on the none-independent identically distributed dataset (i.e., N-IDD-$MNIST$).

4.3 Experimental Results on Security Verification

In the Subsection 3.3, the security verification has also been discussed. Here, the state-of-the-art algorithm DLG [30] is selected as baseline, and the metric Mean Square Error (MSE) is employed to evaluate the performance of our proposed SAM. MSE is an evaluation indicators widely utilized in the researches of regression analysis. Intuitively, the smaller the value of MSE is, the higher the risk of data leakage is. The indicator are defined as follows.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(\hat{Q} - Q)^2 \tag{5}$$

where Q and $\hat{Q}$ respectively denote the real and dummy gradients, and N denotes the number of samples.

In order to make a fair comparison, we follow the experimental environments and the corresponding experimental settings utilized in [30]. The experiments are conducted on dataset $CIFAR$-100 which is also utilized in [30] to verify the security of the proposed SAM. We randomly select two images (i.e., Label 71 and Label 82) as experimental result presentation because of limited space, as shown in Fig. 7 and Fig. 8 respectively.

Fig. 7 shows the restoration processes on Label 71 under algorithms DLG and SAM perform. Apparently, the smaller the value of MSE is, the better the effectiveness of the image restoration is, and the higher the risk of data leakage is. Throughout the whole process that DLG performs, Label 71 has been restored when the number of the iterations reach 100, whereas the image cannot be restored during the process that SAM performs. Fig. 8 shows the restoration processes on Label 82 under algorithms DLG and SAM perform. Similarly, throughout the whole process that DLG performs, Label 82 has been restored when the number of the iterations reach 90, whereas the image cannot be restored during the process that SAM performs. This is because by confusing the gradients in each update operation, MSE between real and dummy gradients randomly changes so that it cannot be minimized.

Therefore, we can conclude that the proposed SAM passes the security verification since it performs excellent on dataset $CIFAR$-100. That is, algorithm SAM greatly reduces the risk of data leakage.

5 Related Work

Our D-FLSPL involves two main components, i.e., distributed learning with privacy-preserving and attack patterns on distributed learning models. Therefore, in the following paragraphs, we will review related work in terms of privacy-preserving and attack patterns that the distributed learning methods involve.

5.1 Privacy-Preserving on Distributed Learning Methods

With the increasing awareness of people on the ownership and security of data as well as user privacy, researches on distributed learning with privacy-preserving have been attracting increasing attention. Existing studies on distributed learning with privacy-preserving mainly utilize differential privacy mechanism and secure multi-party computation. For example, Zhang et al. [28] proposed DEQP2 model to achieve privacy by using differential privacy [28]. Pathak et al. [21] proposed a differential privacy-based global classifier by aggregating classifiers trained locally. Hamm et al. [9] trained a global ϵ-differentially private classifier to transfer the $knowledge$ that the local classifier ensemble contains. As far as we know, DSSGD [22], which is proposed by Shokri and Shmatikov, is the first collaborative learning mechanism that enables multiple parties to jointly learn an accurate neural-network model, without sharing their local data. In addition, DSSGD utilizes the sparse vectors [6] to achieve differential privacy mechanism. Chen et al. [4] demonstrate that the synchronous optimization with backup workers can avoid asynchronous noise while mitigating for the worst stragglers. Konečnỳ et al. [14] and McMahan et al. [18] obtained the global update parameters through averaging the local ones, which performs better than asynchronous mode in terms of communication efficiency and privacy-preserving. Mohassel et al. [20] proposed a novel secure machine learn framework SecureML, where clients distribute their local data among two non-colluding servers who learn models on the joint data using secure multi-party computation. Besides, some encryption-based methods [3, 5] utilize the secure summation protocol with tree topology, homomorphic encryption and the secure aggregation protocol.

5.2 Attack Patterns on Distributed Learning Methods

Inference attacks [23] and reconstruction attacks [1], which aim to obtain the private characteristics and information of data owners, greatly challenge the data security and user privacy. For example, attacks that exploit the differences between the outputs of using the trained samples and those of using the untrained samples [23]. Attacks that employ generative adversarial networks to detect overfitting for distinguishing different data distributions [10]. Therefore, studies on Inference attacks and reconstruction attacks also have been attracting increasing attention. Fredrikson et al. [7] proposed a model inversion attack to recover information from

TABLE 1
The comparisons of O-FL and D-FLSPL on three datasets in terms of "$train\ loss$".

Datasets	Methods	The Numbers of Clients						
		10	50	100	150	200	250	300
MNIST	O-FL	0.00046	0.00079	0.00146	0.00531	0.01228	0.01822	0.02902
	D-FLSPL	0.00053	0.00094	0.00235	0.00784	0.01657	0.02741	0.03516
N-IDD-MNIST	O-FL	0.00149	0.01719	0.04118	0.04387	0.06271	0.06140	0.29502
	D-FLSPL	0.00162	0.01835	0.04793	0.05149	0.06935	0.06627	0.28167
CIFAR-10	O-FL	0.15353	0.12833	0.27847	0.41254	0.52216	0.57343	0.66486
	D-FLSPL	0.15247	0.13637	0.28372	0.41958	0.52495	0.57924	0.66374

TABLE 2
The comparisons of O-FL and D-FLSPL on three datasets in terms of "$test\ accuracy$".

Datasets	Methods	The Numbers of Clients						
		10	50	100	150	200	250	300
MNIST	O-FL	0.9997	0.9994	0.9995	0.9991	0.9987	0.9972	0.9934
	D-FLSPL	0.9989	0.9985	0.9983	0.9982	0.9979	0.9965	0.9928
N-IDD-MNIST	O-FL	0.9993	0.9951	0.9868	0.9855	0.9821	0.9820	0.9784
	D-FLSPL	0.9992	0.9943	0.9861	0.9843	0.9815	0.9814	0.9779
CIFAR-10	O-FL	0.9489	0.9557	0.9064	0.8599	0.8265	0.8044	0.7743
	D-FLSPL	0.9482	0.9549	0.9057	0.8591	0.8261	0.8041	0.7749

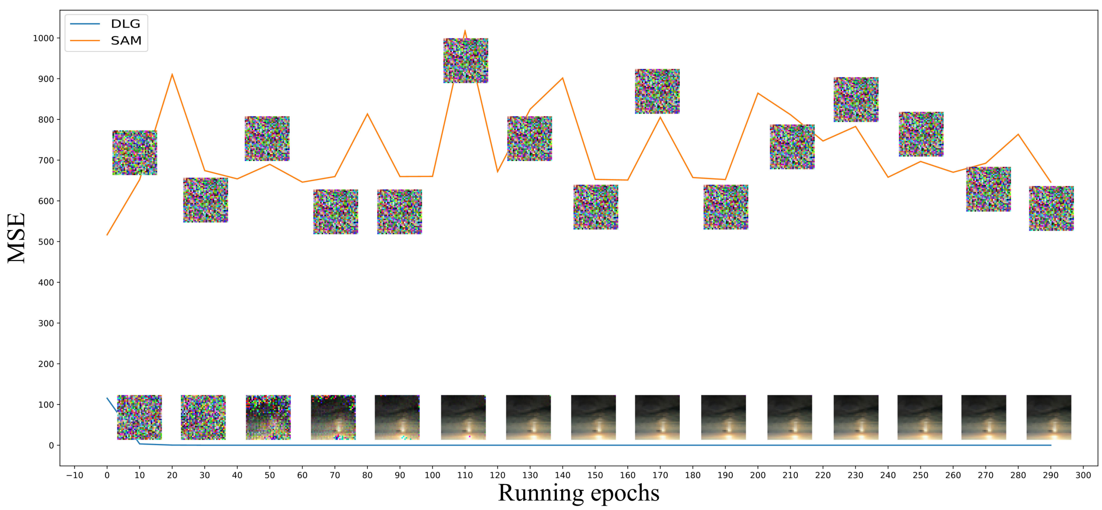


Fig. 7. The experimental comparisons between DLG and SAM on Label 71.

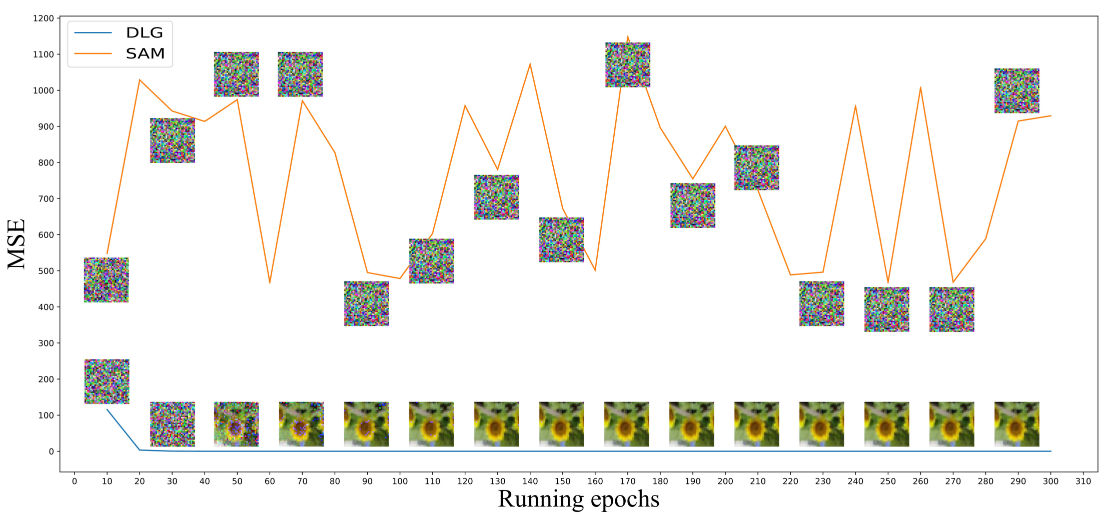


Fig. 8. The experimental comparisons between DLG and SAM on Label 82.

face recognition services by maximizing the confidence value of white noise images.

Melis et al. [19] first proposed an inference attack against collaborative/federated learning. Aono et al. [2] recovered partial private data of clients according to the ratios between training data and gradients sent to the central server, based on an assumption that a malicious server exists. Hitaj et al. [11] reconstructed the private data of some clients based on an assumption that a malicious client exists in federated learning. Wang et al. [24] proposed a novel framework combining generative adversarial networks and multitasking discriminator, namely mGAN-AI. mGAN-AI can simultaneously identify the categories, authenticity of input samples as well as the identities of clients, thus realizing user-level privacy inference attack. Zhu et al. [30] proposed an algorithm that obtains the private information from gradient switching. That is, the server reverse recovers the private data of clients via gradients uploaded.

6 Conclusions and Future Work

This study proposed a novel federal learning-based framework, namely D-FLSPL, to further defend server-level privacy leakage derived from inference attacks and reconstruction attacks. D-FLSPL utilizes secure aggregation mechanism of confusion gradients, namely SAM, to reconstruct the gradient of each client, thus preserving attacks from curious central server and curious client(s). The experimental results showed the correctness and the security of D-FLSPL.

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